

25.2 Potential outcomes and the regression coefficient

Whether a treatment works needs much less than a model of the whole mechanism. Each unit has two possible responses, with and without the treatment, and the effect on that unit is their difference. This language of potential outcomes, due to Neyman (1923) for field trials and extended to observational studies by Rubin (1974), needs no model for the responses, and under randomization it gives exact results. The regression coefficient of a treatment indicator is the natural estimator; its usual standard error is wrong in a specific way.

25.2.1 Potential outcomes

Definition 25.2.1. Potential outcomes and average effects

Consider n units and a binary treatment. Unit i has two **potential outcomes**: $y_i(1)$, the response it would show if treated, and $y_i(0)$, the response it would show if not. With $T_i \in \{0, 1\}$ the treatment it actually receives, the observed response is

$$Y_i = T_i y_i(1) + (1 - T_i) y_i(0). \quad (25.2.1)$$

The **unit effect** is $\tau_i = y_i(1) - y_i(0)$, and the **average treatment effect** of the n units is

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \bar{y}(1) - \bar{y}(0).$$

When the units are instead drawn at random from a population, so that $(Y(0), Y(1), T)$ is a random vector, the population **average treatment effect** is $E(Y(1) - Y(0))$ and the **average effect on the treated** is $E(Y(1) - Y(0) \mid T = 1)$.

Equation Eq. 25.2.1 assumes that unit i 's response depends only on its own treatment (**no interference**, which fails for vaccines) and that there is one version of each treatment: Rubin's *stable unit treatment value assumption*. In a structural causal model, $Y(t)$ is the value of Y , as a function of the disturbances, in the model modified by $\text{do}(T = t)$, so $E Y(t) = E(Y \mid \text{do}(T = t))$. On the event $T = t$, solving the original and the modified assignments in order gives every variable the same value, so $Y = Y(t)$ there, which is Eq. 25.2.1.

For each unit one potential outcome is observed and the other is **counterfactual**, so no unit effect is ever observed (Holland's (1986) fundamental problem of causal inference). Averages can still be estimated from *different* units, if treated and untreated units are comparable.

Proposition 25.2.1. Selection bias

Let $(Y(0), Y(1), T)$ be random with finite means and $0 < \Pr(T = 1) < 1$. Then

$$E(Y \mid T = 1) - E(Y \mid T = 0) = E(Y(1) - Y(0) \mid T = 1) + [E(Y(0) \mid T = 1) - E(Y(0) \mid T = 0)]. \quad (25.2.2)$$

Proof. By Eq. 25.2.1, $E(Y | T = 1) = E(Y(1) | T = 1)$ and $E(Y | T = 0) = E(Y(0) | T = 0)$. Add and subtract $E(Y(0) | T = 1)$. ■

The observed difference is the effect on the treated plus a **selection bias**, the difference between treated and untreated units in the response they would have shown untreated. In Example (Clinic visits and health) heavy users of the clinic would have been less healthy than light users even without their visits. The data on (T, Y) cannot reveal the bias, because $Y(0)$ is never observed for the treated.

25.2.2 Randomization

A **completely randomized experiment** treats n_1 of the n units, every subset of size n_1 being equally likely, and leaves the other $n_0 = n - n_1$ as controls. In Neyman's framework the potential outcomes are fixed numbers and the only randomness is the assignment: no linearity, normality or common variance is assumed, and the units need not be a sample from anything.

Write $\bar{Y}_1$ and $\bar{Y}_0$ for the mean observed responses of the treated and control units, s_1^2 and s_0^2 for their sample variances, and

$$S_t^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i(t) - \bar{y}(t))^2, \quad S_\tau^2 = \frac{1}{n-1} \sum_{i=1}^n (\tau_i - \tau)^2$$

for the variances of the potential outcomes and of the unit effects over the n units. The treated units are a simple random sample from the n units, and the moments of such samples are as follows (compare Lemma 18.4.3).

Lemma 25.2.2. Means of a simple random sample

Let $u_1, \dots, u_N$ and $v_1, \dots, v_N$ be fixed numbers with means $\bar{u}, \bar{v}$ and $S_{uv} = \sum_a (u_a - \bar{u})(v_a - \bar{v}) / (N-1)$. If $\mathcal{S}$ is a simple random sample of m of the indices, $1 \leq m \leq N$, with sample means $\bar{u}_\mathcal{S}$ and $\bar{v}_\mathcal{S}$, then

$$E \bar{u}_\mathcal{S} = \bar{u}, \quad \text{Cov}(\bar{u}_\mathcal{S}, \bar{v}_\mathcal{S}) = \left(\frac{1}{m} - \frac{1}{N} \right) S_{uv}.$$

Proof. Let W_a indicate that $a \in \mathcal{S}$. Then $E W_a = m/N$, and for $a \neq c$, $E W_a W_c = m(m-1) / \{N(N-1)\}$. Hence $\text{Var } W_a = q$ and $\text{Cov}(W_a, W_c) = m(m-1) / \{N(N-1)\} - m^2/N^2 = -q/(N-1)$, where $q = (m/N)(1-m/N)$. Put $\tilde{u}_a = u_a - \bar{u}$ and $\tilde{v}_a = v_a - \bar{v}$. Since $\sum_a W_a = m$, $\bar{u}_\mathcal{S} - \bar{u} = m^{-1} \sum_a W_a \tilde{u}_a$, whose mean is $N^{-1} \sum_a \tilde{u}_a = 0$. For the covariance,

$$m^2 \text{Cov}(\bar{u}_\mathcal{S}, \bar{v}_\mathcal{S}) = \sum_{a,c} \text{Cov}(W_a, W_c) \tilde{u}_a \tilde{v}_c = q \left(\sum_a \tilde{u}_a \tilde{v}_a - \frac{1}{N-1} \sum_{a \neq c} \tilde{u}_a \tilde{v}_c \right).$$

Because $\sum_a \tilde{u}_a = 0$, $\sum_{a \neq c} \tilde{u}_a \tilde{v}_c = -\sum_a \tilde{u}_a \tilde{v}_a$, and the bracket is $\sum_a \tilde{u}_a \tilde{v}_a N / (N-1) = N S_{uv}$. So the covariance is $q N S_{uv} / m^2 = (1/m - 1/N) S_{uv}$. ■

Theorem 25.2.3. Randomization and the difference in means

In a completely randomized experiment with $n_1 \geq 1$ and $n_0 \geq 1$, let $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$.

- (a) $\hat{\tau}$ is unbiased for the average treatment effect: $E \hat{\tau} = \tau$.
- (b) Its variance over the randomization is

$$\text{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_\tau^2}{n}. \quad (25.2.3)$$

- (c) If $n_1, n_0 \geq 2$, the estimator $\hat{V} = s_1^2/n_1 + s_0^2/n_0$ has $E \hat{V} = \text{Var}(\hat{\tau}) + S_\tau^2/n \geq \text{Var}(\hat{\tau})$, with equality iff the unit effects are all equal.
- (d) $\hat{\tau}$ is the least squares coefficient of T in the regression of the observed responses on an intercept and the treatment indicator.
- (e) If instead the units are independent copies of $(Y(0), Y(1), T)$ with T independent of $(Y(0), Y(1))$, $0 < \Pr(T = 1) < 1$ and finite means, then $E(Y | T = 1) - E(Y | T = 0) = E(Y(1) - Y(0))$, and the average effect on the treated equals the average treatment effect.

Proof. Let $\mathcal{S}$ be the set of treated units, a simple random sample of size n_1 . Then $\bar{Y}_1$ is the sample mean of the $y_i(1)$ over $\mathcal{S}$, and $\bar{Y}_0$ is the mean of the $y_i(0)$ over the complement.

- (a) The complement of $\mathcal{S}$ is a simple random sample of size n_0 . By Lemma 25.2.2, $E \bar{Y}_1 = \bar{y}(1)$ and $E \bar{Y}_0 = \bar{y}(0)$.
- (b) Let $\bar{y}_{\mathcal{S}}(0)$ be the mean of the $y_i(0)$ over $\mathcal{S}$. The control mean is $\bar{Y}_0 = (n\bar{y}(0) - n_1\bar{y}_{\mathcal{S}}(0))/n_0$, so

$$\hat{\tau} = \bar{u}_{\mathcal{S}} - \frac{n}{n_0}\bar{y}(0), \quad u_i = y_i(1) + \frac{n_1}{n_0}y_i(0). \quad (25.2.4)$$

By Lemma 25.2.2, $\text{Var}(\hat{\tau}) = (1/n_1 - 1/n)S_u^2 = \{n_0/(nn_1)\}S_u^2$. Expanding, $S_u^2 = S_1^2 + (n_1/n_0)^2S_0^2 + 2(n_1/n_0)S_{10}$, where S_{10} is the covariance of the two potential outcomes over the units, and $S_\tau^2 = S_1^2 + S_0^2 - 2S_{10}$. Eliminating S_{10} ,

$$S_u^2 = \frac{n}{n_0}S_1^2 + \frac{nn_1}{n_0^2}S_0^2 - \frac{n_1}{n_0}S_\tau^2,$$

and multiplying by $n_0/(nn_1)$ gives Eq. 25.2.3.

- (c) The treated responses are a simple random sample of $n_1 \geq 2$ of the numbers $y_i(1)$, so $E s_1^2 = S_1^2$ by Lemma 18.4.3; likewise $E s_0^2 = S_0^2$. Hence $E \hat{V} = S_1^2/n_1 + S_0^2/n_0$, which exceeds Eq. 25.2.3 by S_τ^2/n . This is zero iff $\tau_i = \tau$ for every i .
- (d) This is Proposition 5.4.4 with two groups and control as the reference.
- (e) By Eq. 25.2.1 and independence, $E(Y | T = 1) = E(Y(1) | T = 1) = E Y(1)$, and likewise for $T = 0$. The same independence gives $E(Y(1) - Y(0) | T = 1) = E(Y(1) - Y(0))$.

■

Part (e) shows randomization removing the selection bias of Eq. 25.2.2; part (a) needs no sampling model at all. The last term of (b) cannot be estimated, because it depends on how $y_i(1)$ is paired with $y_i(0)$ within units, which no unit reveals. Part (c) accepts a conservative

answer, exact when the treatment adds the same amount to every unit.

Example. All randomizations of twenty units

A synthetic population of $n = 20$ units has $\tau = 2.100$, $S_1^2 = 12.129$, $S_0^2 = 5.678$ and $S_\tau^2 = 1.698$; the effect grows with $y_i(0)$. With $n_1 = 8$ there are 125970 assignments, few enough to enumerate. Over all of them $\hat{\tau}$ has mean 2.100 and variance 1.9043, which is Eq. 25.2.3 to every digit. Neyman's $\hat{V}$ has mean 1.9892, too large by $S_\tau^2/n = 0.0849$. The pooled least squares variance estimate averages 1.7055, below the true variance, because it gives the more variable treated arm the weight of the smaller arm (Exercise (B1)). Figure 25.2.1 shows the distribution of $\hat{\tau}$.

```
import itertools
import numpy as np

rng = np.random.default_rng(2502)
n, n1 = 20, 8
n0 = n - n1
y0 = np.round(rng.normal(10, 3, size=n), 1)           # potential outcome
      , control
y1 = np.round(y0 + 2 + 0.5 * (y0 - 10) + rng.normal(0, 1, n), 1) # potential
      outcome, treated
tau = np.mean(y1 - y0)                                # the sample
      average effect
S1, S0, St = np.var(y1, ddof=1), np.var(y0, ddof=1), np.var(y1 - y0, ddof=1)
print(f"tau = {tau:.3f}; S1^2 = {S1:.3f}, S0^2 = {S0:.3f}, S_tau^2 = {St:.3f}")
```

[Run this code online](#)

```
idx = np.array(list(itertools.combinations(range(n), n1))) # every possible
      treated set
treated = np.zeros((len(idx), n), dtype=bool)
treated[np.arange(len(idx))[:, None], idx] = True        # one row per
      assignment
m1, m0 = (treated @ y1) / n1, (~treated @ y0) / n0       # arm means, one
      row per assignment
diff = m1 - m0                                           # difference in
      means
s1 = (treated @ y1 ** 2 - n1 * m1 ** 2) / (n1 - 1)       # within-arm
      variances
s0 = (~treated @ y0 ** 2 - n0 * m0 ** 2) / (n0 - 1)
V_hat = s1 / n1 + s0 / n0                               # Neyman's
      variance estimate
V_neyman = S1 / n1 + S0 / n0 - St / n
print(f"{len(diff)} assignments")
print(f"mean of difference in means {diff.mean():.3f} (tau = {tau:.3f})")
print(f"variance over assignments {diff.var():.4f} (Neyman's formula {V_neyman:.4f})")
print(f"mean of variance estimate {V_hat.mean():.4f} (excess S_tau^2/n = {St / n:.4f})")
```

[Run this code online](#)

Finite-population central limit theorems (Hájek 1960; Li and Ding 2017), whose proofs are beyond this chapter, make $(\hat{\tau} - \tau)/\sqrt{\text{Var } \hat{\tau}}$ asymptotically standard normal under conditions on

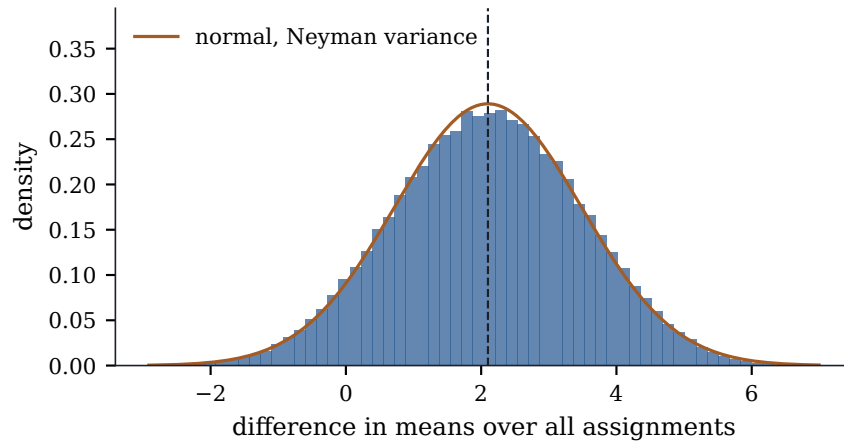


Figure 25.2.1: The randomization distribution of the difference in means in [Example \(All randomizations of twenty units\)](#), over all 125970 assignments of 8 of the 20 units to treatment. The dashed line is τ ; the curve is the normal density with Neyman’s variance [Eq. 25.2.3](#).

the potential outcomes, so $\hat{\tau} \pm 1.96\sqrt{\hat{V}}$ has coverage at least 95% in the limit. In the example, with twenty units, it covers τ for a fraction 0.931 of the assignments.

25.2.3 What the regression coefficient estimates

By part (d), the least squares coefficient of T is unbiased for the average effect because of the design, not because the model $E(Y_i) = \beta_0 + \beta_1 T_i$ is true: its “errors” are not random draws and have different variances in the two arms. What the model gets wrong is the standard error. The pooled estimate of $\sigma^2(1/n_1 + 1/n_0)$ (Theorem 5.5.1) can be too small or too large when $n_1 \neq n_0$ ([Exercise \(B1\)](#)), while Neyman’s $\hat{V}$ is exactly the HC2 sandwich estimate of [Chapter 21 \(Exercise \(B2\), Theorem 21.4.2\)](#). Covariates measured before randomization can be added to this regression to gain precision; [Section 25.6](#) shows how, and finds a surprise.

Remark (Fisher’s sharp null). Fisher (1935) tested the *sharp* null $y_i(1) = y_i(0)$ for every unit, under which the randomization distribution of any statistic is known exactly. [Chapter 23](#) develops these permutation tests (Theorem 23.4.2).

25.2.4 Exercises

25.2.4.1 A. Check your understanding

Exercise. A1

Four units have potential outcomes $(y_i(0), y_i(1)) = (1, 4), (3, 3), (5, 9), (7, 8)$. Two are treated by complete randomization. List the six possible values of $\hat{\tau}$, and verify [Theorem 25.2.3\(a\)](#) and (b) directly.

Solution. The unit effects are 3, 0, 4, 1, so $\tau = 2$. For the treated pairs $\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}$ the treated means are 3.5, 6.5, 6, 6, 5.5, 8.5 and the control means 6, 5, 4, 4, 3, 2, giving $\hat{\tau} = -2.5, 1.5, 2, 2, 2.5, 6.5$. Their mean is $12/6 = 2 = \tau$. Their squared deviations from 2 sum to $20.25 + 0.25 + 0 + 0 + 0.25 + 20.25 = 41$, so $\text{Var } \hat{\tau} = 41/6$. The formula: $S_1^2 = 26/3$,

$$S_0^2 = 20/3, S_\tau^2 = 10/3, \text{ and } (26/3)/2 + (20/3)/2 - (10/3)/4 = 46/6 - 5/6 = 41/6.$$

25.2.4.2 B. Practice

Exercise. B1

Let $s_p^2 = \{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2\}/(n - 2)$ and $\hat{V}_{OLS} = s_p^2(1/n_1 + 1/n_0)$, the least squares variance estimate for the coefficient of T . Show that $\hat{V}_{OLS} = \hat{V}$ when $n_1 = n_0$, and that in general

$$E \hat{V}_{OLS} - E \hat{V} = \frac{(n - 1)(n_1 - n_0)(S_1^2 - S_0^2)}{n_1 n_0 (n - 2)}.$$

Conclude that when the smaller arm has the more variable potential outcomes, the least squares variance estimate is smaller on average than Neyman's, and it can be smaller than the true variance.

Solution. If $n_1 = n_0 = m$, then $s_p^2 = (s_1^2 + s_0^2)/2$ and $\hat{V}_{OLS} = s_p^2 \cdot 2/m = \hat{V}$. In general $E s_t^2 = S_t^2$ (Lemma 18.4.3), so $E \hat{V}_{OLS} = n\{(n_1 - 1)S_1^2 + (n_0 - 1)S_0^2\}/\{n_1 n_0 (n - 2)\}$. The coefficient of S_1^2 in $E \hat{V}_{OLS} - E \hat{V}$ is

$$\frac{n(n_1 - 1) - n_0(n - 2)}{n_1 n_0 (n - 2)} = \frac{n(n_1 - n_0) - (n_1 - n_0)}{n_1 n_0 (n - 2)} = \frac{(n - 1)(n_1 - n_0)}{n_1 n_0 (n - 2)},$$

using $n = n_1 + n_0$; by symmetry the coefficient of S_0^2 is its negative. If $n_1 < n_0$ and $S_1^2 > S_0^2$, the difference is negative. Since $E \hat{V}$ exceeds $\text{Var } \hat{\tau}$ only by S_τ^2/n , the least squares estimate underestimates the variance on average whenever the negative difference is larger than that. In [Example \(All randomizations of twenty units\)](#), $E \hat{V}_{OLS} = 1.7055$ while $\text{Var } \hat{\tau} = 1.9043$.

Exercise. B2

In the regression of the observed Y_i on $\mathbf{1}$ and the treatment indicator, the leverages are $h_{ii} = 1/n_1$ for treated units and $1/n_0$ for controls (Proposition 6.8.1). The HC2 estimate of $\text{Cov}(\hat{\beta})$ is $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{diag}(\hat{e}_i^2/(1 - h_{ii})) \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}$. Show that its entry for the coefficient of T is exactly Neyman's $\hat{V}$.

Solution. Reparameterize with the two arm indicators as columns, which changes neither the fit nor the estimate of the difference of the arm means (Proposition 5.7.1). Then $\mathbf{X}^\top \mathbf{X} = \text{diag}(n_1, n_0)$, the residuals are $Y_i - \bar{Y}_1$ and $Y_i - \bar{Y}_0$, and the sandwich is diagonal with entries $n_1^{-2} \sum_{\text{treated}} \hat{e}_i^2/(1 - 1/n_1) = s_1^2/n_1$ and similarly s_0^2/n_0 . The variance of the difference of the two coefficients is their sum, $\hat{V}$.

25.2.4.3 C. Going deeper

Exercise. C1

Show that $S_\tau^2 \geq (S_1 - S_0)^2$, and hence that

$$\text{Var}(\hat{\tau}) \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{(S_1 - S_0)^2}{n},$$

a bound that is attained when $y_i(1)$ is an increasing affine function of $y_i(0)$. Propose a conservative variance estimator that is never larger than $\hat{V}$.